

ENDOCRINE PHYSIOLOGY OBJECTIVES

PARATHYROID HORMONE AND VITAMIN D: Hormonal Regulation of Ca^{++} & PO_4 .

1. Know the general distribution & functions of calcium in the body.
2. Know the compartment of body calcium directly "sensed" by calcium regulating hormones
3. Know the major hormones that regulate calcium & phosphate homeostasis and their principal target tissues.
4. Know the basic hormone properties of parathyroid hormone and compare and contrast with other endocrine hormones.
5. Know the specific actions of PTH in the kidneys including the specific sites of action along the renal tubules related to regulation of calcium and phosphate homeostasis.
6. Understand why PTH has only limited capacity to reabsorb calcium from renal tubules.
7. Rationalize the physiological benefit of PTH's action of increasing PO_4 excretion in relation to calcium homeostasis.
8. Know the specific role of PTH in the regulation of vitamin D biosynthesis.
9. Review the basic structure and histology of cortical bone as it relates to the calcium & phosphate regulatory action of hormones.
10. Know what hydroxyapatite is and its relationship with soluble bone salts and bone fluid.
11. Explain bone remodeling in terms of bone synthesis and resorption and in relation to the roles of osteoclasts, osteoblasts, and osteocytes.
12. Know the major factors that regulate PTH secretion.
13. Describe the events that occur during the rapid and slow phases of PTH actions in bone in response to acute and chronic hypocalcemia.
14. Know the pathophysiological manifestations resulting from primary hyperparathyroidism and hypoparathyroidism and relate to the physiological actions of PTH.
15. Be able to predict laboratory diagnostic parameters associated with hyperparathyroidism and hypoparathyroidism
16. Know the basic hormone properties of Vitamin D and compare and contrast with other steroid hormones as well as protein hormones.
17. Explain the biosynthesis of $1,25(\text{OH})_2\text{cholecalciferol}$ from skin to the liver to kidneys.
18. Know the physiological actions of vitamin D that result in increased calcium and phosphate absorption from the intestines.
19. Know the factors that regulate vitamin D biosynthesis.
20. Know the general clinical manifestations that occur as a result of rickets or osteomalacia and be able to relate these symptoms to the normal physiological actions of vitamin D
21. Know the basic hormone properties of calcitonin
22. What are the basic actions of calcitonin in relation to calcium regulation?
23. Be able to describe the dynamics and sequence of events during the integrated hormonal response (PTH, Vitamin D & calcitonin) to hypocalcemia and hypercalcemia.